

Circuit Simulation

Test circuits before you build them · Equigrow short notes

Analyses

Analysis	Shows
.op	DC operating point (voltages/currents)
.tran	behaviour over time
.ac	frequency response (Bode plot)
.dc	sweep a source

Workflow

- Draw the schematic, add a ground, set source values, choose an analysis, run, and probe nodes.
- Compare results with hand calculations (Ohm's law, KVL, KCL).

Core formulas

- $V = IR$ · $P = VI$ · RC time constant $\tau = RC$ · cut-off frequency $f = 1/(2\pi RC)$.